

Database Systems

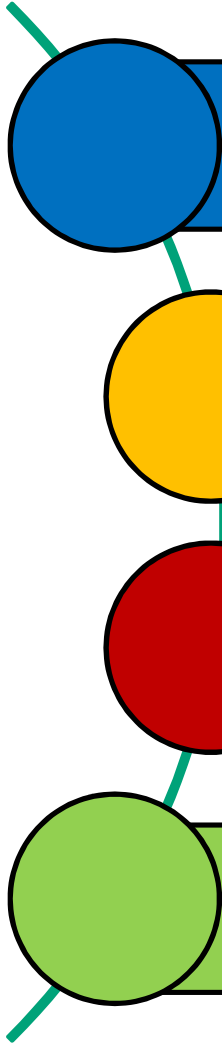
CSE 303

Lecture 06: SQL

2022

Subquery

Outline



What is a Subquery

Subquery in WHERE clause

>ALL, >ANY, >=ALL, <=ANY etc.

Subquery in FROM clause

What is Sub-query?

Queries within query

Query as a part of another query

2

SELECT <attributes>

FROM <one or more relations>

1

WHERE <conditions>

Sub-query in WHERE clause

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

Sub-query in WHERE clause

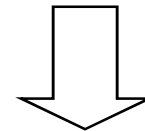
Find the employee(s) with the highest paid salary

First thought

```
SELECT id  
FROM Employee  
WHERE salary = 75000
```

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000



id	dept	salary
5004	CSE	75,000

Sub-query in WHERE clause

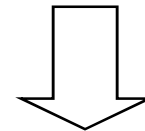
Find the employee(s) with the highest paid salary

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

Suppose you are
not allowed to use
any aggregate
function??

```
SELECT id
FROM Employee
WHERE salary = (SELECT MAX(salary)
                FROM Employee)
```



id	dept	salary
5004	CSE	75,000

Sub-query in WHERE clause

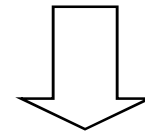
Find the employee with the highest paid salary

> ALL	>= ALL
< ALL	<= ALL
> ANY	>= ANY
< ANY	<= ANY

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

```
SELECT id
FROM Employee
WHERE salary >= ALL (SELECT salary
                     FROM Employee)
```



id	dept	salary
5004	CSE	75,000

Correlated sub queries in the WHERE clause

Find the employee(s) with the highest paid salary

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

```
SELECT id
FROM Employee
WHERE salary > ALL (SELECT salary
                    FROM Employee
                    WHERE salary < ANY (SELECT salary
                                        FROM Employee))
)
```

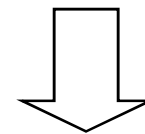
Correlated sub queries in the WHERE clause

Find the highest paid employee(s) in each department

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	55,000
5004	CSE	75,000

```
SELECT id, dept
FROM Employee
WHERE (dept, salary) IN (SELECT dept, MAX(salary)
                        FROM Employee
                        GROUP BY dept);
```



id	dept	salary
5002	EEE	55,000
5004	CSE	75,000

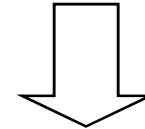
Correlated sub queries in the WHERE clause

Find the employees that earn above average

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

```
SELECT *  
FROM Employee  
WHERE salary > 60000
```



id	dept	salary
5004	CSE	75,000

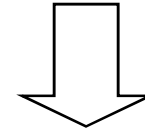
Correlated sub queries in the WHERE clause

Find the employees that earn above average

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

```
SELECT *  
FROM Employee  
WHERE salary > (SELECT AVG(salary)  
                FROM Employee)
```



id	dept	salary
5004	CSE	75,000

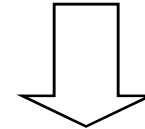
Correlated sub queries in the WHERE clause

Find the employees that
earn above average of
the employees within
respective department

Employee

id	dept	salary
5001	EEE	50,000
5002	EEE	55,000
5003	CSE	60,000
5004	CSE	75,000

```
SELECT *  
FROM Employee s  
WHERE salary > (SELECT AVG(salary)  
                FROM Employee t  
                WHERE s.dept = t.dept);
```



id	dept	salary
5002	EEE	55,000
5004	CSE	75,000

EXISTS and NOT EXISTS

```
SELECT <attributes>  
FROM   <one or more relations>  
WHERE  EXISTS (sub-query)
```



Empty
FALSE

Not Empty
TRUE

Find all manufacturers that make both PC and Laptop

PC

model	maker
1001	A
1002	B
1003	C

Laptop

model	maker
2001	A
2002	B
2003	D

```
SELECT DISTINCT maker
FROM PC
WHERE EXISTS (SELECT *
               FROM Laptop
               WHERE PC.maker =
                   Laptop.maker)
```



maker
A
B

Find all manufacturers that make PCs but not Laptops

PC

model	maker
1001	A
1002	B
1003	C

Laptop

model	maker
2001	A
2002	B
2003	D

```
SELECT DISTINCT maker
FROM PC
WHERE NOT EXISTS (SELECT *
                  FROM Laptop
                  WHERE PC.maker =
                    Laptop.maker)
```



maker
C

Practice correlated sub queries

Find the PC model with
maximum processing speed
using and without using
aggregate function

```
SELECT model
FROM PC
WHERE speed = (SELECT MAX(speed)
                FROM PC);
```

```
SELECT model
FROM PC
WHERE speed >= ALL (SELECT speed
                    FROM PC)
```

PC

MODEL	SPEED	RAM	HD	PRICE
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

Correlated sub queries in the FROM clause

Find manufacturers of at least two different computers
(PC's or laptops) with speeds of at least 2.0

Maker	Model	Type
A	1001	pc
A	1002	pc
A	1003	pc
A	2004	laptop
A	2005	laptop
A	2006	laptop
B	1004	pc
B	1005	pc
B	1006	pc
B	2007	laptop
C	1007	pc
...	...	...
....	...	...

MODEL	SPEED	RAM	HD	PRICE
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
...		...	...	

PC

MODEL	SPEED	RAM	HD	SCREEN	PRICE
2001	2	2048	240	20.1	3673
2002	1.73	1024	80	17	949
2003	1.8	512	60	15.4	549
2004	2	512	60	13.3	1150
....					

Laptop

Product

Find manufacturers of at least two different computers
(PC's or laptops) with speeds of at least 2.0

```
SELECT maker
FROM (SELECT model, maker
      FROM Product NATURAL JOIN PC
      WHERE speed >= 2
      UNION
      SELECT model, maker
      FROM Product NATURAL JOIN Laptop
      WHERE speed >= 2)
GROUP BY maker
HAVING COUNT(model) >= 2;
```

Find manufacturer of at least two
different PCs with speed of at least 2.0

```
SELECT maker  
FROM   Product, PC  
WHERE  Product.model = PC.model  
AND    speed >= 2.0  
GROUP BY maker  
HAVING COUNT(PC.model) >= 2
```

Find manufacturer of at least two
different Laptops with speed of at least 2.0

```
SELECT maker  
FROM   Product, Laptop  
WHERE  Product.model = Laptop.model  
AND    speed >= 2.0  
GROUP BY maker  
HAVING COUNT(Laptop.model) >= 2
```

Find manufacturers of at least two different computers
(PC's or laptops) with speeds of at least 2.0

Think combined

```
SELECT maker
FROM   Product, (SELECT model, speed FROM PC
                  UNION
                  SELECT model, speed FROM Laptop) T
WHERE  Product.model = T.model
AND    speed >= 2.0
GROUP BY maker
HAVING COUNT(T.model) >= 2
```